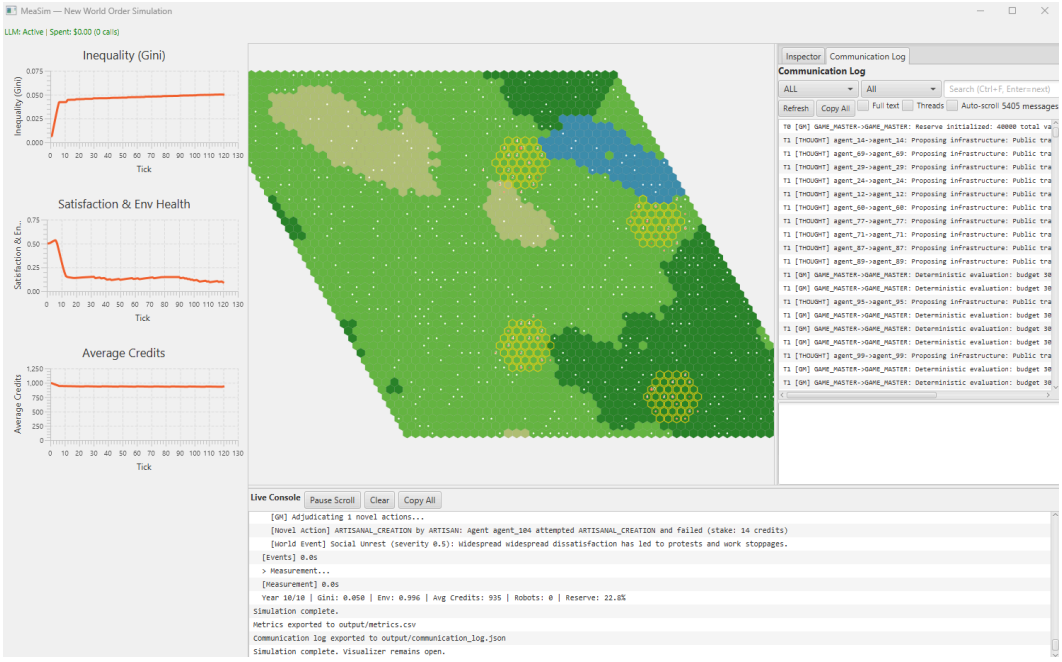
MeaSim: Agent-Based Simulation of Post-Labor Economic Systems



Policy Challenge and Significance

Capitalism distributes money primarily through employment. AI and robotics are making human labor optional across increasing sectors of the economy. This creates a structural contradiction: the more productive our technology becomes, the less capable our economic system is of distributing the wealth that technology creates.

This is not a future problem. Automation is already displacing workers faster than new job categories emerge. The standard policy responses — retraining programs, universal basic income proposals, regulation of AI deployment — are designed as patches to an existing system, not as coherent alternatives. None of them address the fundamental design flaw: an economic system that requires human labor scarcity to function cannot survive human labor abundance.

The challenge is not the lack of proposed alternatives. It is a lack of testing. Economic policy is uniquely resistant to empirical validation. You cannot A/B test a national economy. You cannot roll back a failed monetary policy like a bad deployment. The result is that economic systems are designed through ideology and implemented through crisis, with no engineering discipline applied to their architecture.

We need a way to stress-test economic systems before deploying them — the same way we stress-test software before releasing it to production. Any economic system that cannot survive adversarial agents in simulation will not survive them in reality.

The significance is immediate. AI capability is advancing on a timeline measured in months. Economic policy operates on a timeline measured in decades. If we do not develop and validate alternative economic frameworks now, we will be forced to improvise them during a crisis.

Proposed Policy Solution

We propose MEAS (Meritocratic Externality-Aware Economic System), a redesigned economic framework, and MeaSim, a full agent-based simulator built to test it.

MEAS preserves capitalism's core strength — incentive alignment through gamification — while fixing its structural flaws. It operates on five scoring axes applied to all economic transactions:

• Environmental Footprint (EF): Pollution and resource depletion cost more. Clean operations keep more revenue.
• Commons Contribution (CC): Activity benefiting underserved populations earns a scoring bonus.
• Labor Displacement (LD): Automation revenue is partially diverted to a UBI pool, scaling the safety net to the displacement.
• Resource Concentration (RC): Wealth accumulation faces progressive friction. No hard caps, but decelerating returns.
• Economic Productivity (EP): Value creation is rewarded. Innovation and efficiency still pay.

These modifiers apply automatically to all credit flows — protocol-level rules, not taxes.

MeaSim tests this with 200 autonomous AI agents (Claude Sonnet 4.6), each assigned one of 18 adversarial archetypes (Exploiter, Free Rider, Accumulator, Speculator, etc.) designed to stress-test failure modes. A Game Master LLM acts as the physics engine — evaluating infrastructure proposals, setting material costs, enforcing technological consistency. Agents communicate, negotiate, form contracts, and build institutions through LLM reasoning, not hardcoded behaviors.

The simulator runs both MEAS-enabled and pure capitalism modes with identical conditions, producing directly comparable metrics: Gini coefficient, satisfaction distribution, environmental health, and entrepreneurship rates over 50 simulated years.

Potential Implementation Pathway

Phase 1 — Simulation Validation (Current): The simulator is built and functional. We need API credits to run comprehensive 50-year simulations under both MEAS and capitalism conditions across multiple random seeds. Each full run with 200 agents (>Dunbar’s number) costs approximately $6,000 in Anthropic API credits. We need 5-10 runs minimum to establish statistical significance: varying agent counts, archetype distributions, and MEAS parameter settings.

Phase 2 — Formula Refinement: Simulation results will reveal where MEAS scoring curves produce perverse incentives, where agents find exploitable loopholes, and where UBI funding is insufficient. The scoring formulas are designed to be iteratively adjusted — propose, simulate, find bugs, fix, repeat.

Phase 3 — Open Source Community Testing: The project is released open source under Apache 2.0. Contributors can run their own simulations, identify flaws, propose formula improvements, and play as agents using the built-in Player Mode.

Phase 4 — Controlled Real-World Pilot: Once the simulation produces a stable, exploitation-resistant parameter set, the MEAS framework could be piloted in a bounded economic environment — a special economic zone, a digital marketplace, or a DAO governance structure.

Phase 5 — Policy Recommendation: Simulation data, formula specifications, and pilot results packaged as a concrete policy recommendation. The deliverable is tested, versioned, open-source economic software with empirical performance data.

Expected Impact and Relevant Metrics

The primary deliverable is empirical comparison data between MEAS and unregulated capitalism under identical simulated conditions. Key metrics tracked per tick across 50 simulated years:

- Gini coefficient: Does MEAS reduce wealth inequality without suppressing productivity?
- Agent satisfaction: What percentage of agents achieve sustainable subsistence?
- Environmental health: Does EF scoring prevent commons degradation?
- Infrastructure creation: Do agents build more public goods under MEAS incentives?
- Exploitation resilience: Can adversarial archetypes game the scoring system?
- UBI adequacy: Does LD diversion fund sufficient UBI as automation increases?
- Economic velocity: Do MEAS modifiers slow transaction volume?

Secondary impact: MeaSim becomes a reusable tool for testing any economic system design. The open source release invites global participation in refining the MEAS specification through empirical simulation.

Brief Bio and Relevant Expertise

Jack Zante Hays is an AI Specialist and Senior Software Engineer at PayPal on the Developer Productivity Engineering — DevOps AI Tools team. He oversees AI coding tools (Claude Code, Cursor, GitHub Copilot) for PayPal's engineering organization and applies AI to automate software engineering at scale.

In 2023, he automated $30M of technical debt in a single week, becoming the first named inventor of the PayPal patent "Recursive Artificial Intelligence Code Fix Circuit." He is currently targeting billions of dollars in technical debt remediation per month through AI automation.

His direct experience automating human labor at industrial scale — and observing the economic implications — motivated the design of MEAS and the construction of MeaSim. The project applies the same engineering discipline used in production software systems to economic system design.

GitHub: github.com/ZMobile/NewWorldOrderSimulation